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Regulatory policy outputs and impacts: Exploring a complex relationship

Christoph Knill, Kai Schulze, and Jale Tosun

Department of Politics and Public Administration, University of Konstanz, Konstanz, Germany

Abstract

In this article, we pursue the objective of empirically testing the extent to which changes in environmental policy outputs can explain changes in environmental impacts. Previously, systematic testing of this relationship was hampered by the lack of a compelling measurement of changes in regulatory policy outputs. To remedy this, we present a novel approach to the measurement of events of regulatory output change. We illustrate our concept by employing data on changes in clean air regulations in 24 advanced democracies from 1976 to 2003. In a next step, we explore the extent to which changes in clean air regulations can account for changes in air pollutant emissions. The empirical analysis suggests that changes in clean air regulations cannot be unconditionally associated with changing intensities of air pollutant emission. We deem these results to have far-reaching implications for the study of regulatory policy change.

Keywords: clean air regulations, impact data, measurement concept, output data, regulatory policy change.

1. Introduction

Types of regulatory policy have become a popular subject of macro-comparative social science research. In environmental policy, empirical studies interested in explaining variation across countries and over time have typically relied on various kinds of regulatory impact data, such as pollutant emissions and pollution levels (see e.g. Neumayer 2003; Wälti 2004; Roller 2005; Perkins & Neumayer 2008; Bernauer & Koubi 2009). In recent years, however, a growing number of studies have suggested using regulatory output data, for example, compiled from laws and decrees, in order to assess more directly the factors underlying the actors' decisionmaking behavior (see e.g. Givel 2006; Holzinger *et al.* 2008, 2011; Liefferink *et al.* 2009). Following the public policy literature, there should, in principle, be a relationship between regulatory policy outputs and impacts as they both refer to different phases of the same policymaking process. To date, only a small number of studies concentrating on one or a few states have investigated the nature of this supposed relationship empirically (see e.g. Earnhart & Lizal 2008; Koutalakis *et al.* 2010), and even less attention has been paid to how changes in regulatory policy outputs are interrelated to changes in regulatory policy impacts.

Correspondence: Jale Tosun, Politics and Public Administration, University of Konstanz, Box D91, 78457 Konstanz, Germany. Email: jale.tosun@uni-konstanz.de

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It is our goal in this article to empirically illustrate the nature of the relationship between changes in regulatory policy outputs (our focal independent variable) and impacts (our dependent variable) for a larger number of states in order to contribute to the small but growing literature on policy change (see e.g. Radaelli 1997; Richardson 2000; John 2003; Jordana & Sancho 2005; Baumgartner *et al*. 2009; Baumgartner & Jones 2009; Capano & Howlett 2009; Howlett & Cashore 2009; Howlett & Joshi-Koop 2011; Sotirov & Memmler 2012). So far, the implementation of macro-comparative analyses of this relationship has been hampered by the difficulties of measuring changes in regulatory policy outputs. Thus, we present a novel approach to the assessment of this variable, which allows for grasping the complexity of changes in regulatory policy outputs and is yet feasible. The issue of measurement is significant, as without an accurate measurement of changes in regulatory policy outputs we cannot determine their "on the ground" effects, which limits our understanding of how government policies work.

The novel measurement approach is applied to a regulatory policy area that has received considerable scholarly attention, namely environmental protection. More specifically, we focus on the regulation of air pollution and whether changes in policy outputs are reflected in policy impacts as measured by intensities of pollutant emissions. For this purpose, we compile a new data set including changes in clean air regulations in 24 Organisation for Economic Co-operation and Development (OECD) countries over a period of almost three decades (1976–2003).[1] Our estimation results indicate that changes in clean air regulations cannot be systematically related to changes in major air pollutant emissions, thus suggesting that the relationship between adopted regulations and goal attainment needs more careful consideration in future research (see also Levi-Faur 2011).

The remainder of this article is structured as follows: we first introduce our measurement of changes in regulatory policy output in abstract terms, and then illustrate it in a more practical way using the data we coded for assessing changes in clean air regulations. Subsequently, we turn to the measurement of our dependent and other independent variables, before we test the relationship between regulations and air pollutant emissions by applying standard techniques for pooled time-series cross-section data. In the final section we summarize our main findings and present some concluding remarks.

2. Measurement of changes in regulatory policy outputs

To assess the relationship between regulatory policy outputs and impacts we must find appropriate measures for both concepts. In the field of clean air policy, cross-national impact data, like emission levels, are compiled and published by international organizations like the OECD or the World Bank (Wälti 2004). Thus, we do not face significant data supply problems regarding our dependent variable. In marked contrast, comparative output data for clean air regulations, relating, for example, to the rules laid down in laws and decrees, are not readily available. One reason for this absence is that measuring regulatory policy outputs is a challenging task that requires making several coding decisions. Another reason is that the assessment of regulations is resource intensive, for relevant national legal or administrative documents must be gathered and coded. Acknowledging these challenges, we have developed a coding strategy for the assessment of changes in regulatory policy outputs that is comprehensive and yet feasible to implement.

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2.1. Dimensions of change in regulatory policy outputs

Relying on output data allows for a detailed and multidimensional assessment of changes in regulatory policy over time. These can, for instance, refer to basic policy principles or paradigms, the policy instruments, or the concrete settings of these instruments (Hall 1993). Further, data on environmental policy outputs allow for determining the direction of policy change, that is, whether a policy decision increases or lowers regulatory standards. Indeed, the relevant literature often neglects that change is not unidirectional, but can go in two ways, that is, policy expansion and policy dismantling (Capano 2009; Knill *et al.* 2009).

How can we capture regulatory policy change in both its complexity and its inherent direction on the basis of regulatory outputs? In our view, the most compelling measurement approach has been proposed by Hall (1993), on whose work we strongly rely here. In addition, we rely on the conceptual work of Cashore and Howlett (2007) and Howlett and Cashore (2009), who also tried to refine and make Hall's measurement approach quantifiable (for another application, see e.g. Henstra 2011).

On a first level we analyze whether a regulation for a specific target, for example, the quality of drinking water, is in place or not. The second category includes the instruments used to realize the respective regulatory goals, such as emission standards or environmental taxes. The third category of regulatory policy change corresponds to the concrete setting or calibration of the applied instruments. The latter may manifest in emission limits for a certain pollutant, as well as their scope of application; that is, the individuals, organizations, or activities targeted by a specific instrument. Note that by additionally distinguishing the scope category, our typology goes beyond the original scheme put forward by Hall (1993). In our view, this is a necessary complement in order to more accurately assess events of regulatory policy change.

Our coding scheme further extends the original taxonomy by distinguishing between two directions of policy change, namely expansion and dismantling. For example, expansion occurs if a new pollutant becomes subject to regulatory activity. Dismantling, by contrast, would occur if a given pollutant were not regulated anymore. The number of regulatory instruments increases if a new measure to curb emissions of a certain pollutant, for example, a tax, is introduced, and decreases if a measure is abolished. Policymakers can also increase or decrease the concrete calibration of regulatory instruments. For example, the specific levels of a tax, as well as the target group of a tax, can be either increased or decreased. We interpret the first case as an event of expansion and the latter as an event of dismantling. Table 1 illustrates how regulatory policy changes – consisting of policy expansion and dismantling – can be measured along these three dimensions.

2.2. Density change and intensity change

Even when relying on the above-mentioned analytical categories, we still have to clarify how we conceptualize the magnitude and direction of policy change. For this purpose, we distinguish between two basic dimensions: density change and intensity change. Density change concerns changes in the degree of regulatory penetration and internal differentiation of a policy field, and explores how the numbers of policies or instruments change over time. In a complementary vein, intensity change relates to changes in the stringency of the policy instruments, including the levels of regulatory standards, for example, emission limits, and their scope of application, for example, specific industry branches.

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Table 1 Measurement of regulatory policy expansion and dismantling

Dimension of regulation	Expansion	Dismantling
Presence of regulation	Introduction or addition of a new regulation	Dismantling of an existing regulation
Regulatory instruments	The number of instruments increases, e.g. information-based instruments are adopted	The number of instruments decreases, e.g. market-based instruments are abolished
Regulatory calibration: Instrument levels and scopes	Tightening regulatory standards or increasing the target group, e.g. by lowering emission limits	Loosening regulatory standards or decreasing the target group, e.g. by increasing emission limits

Table 2 Dimensions and indicators of regulatory policy change

Change dimension	Sub-dimensions	Indicators
Density change	Regulatory target density	Development of policies over time (difference between number of adopted and abolished policies)
	Regulatory instrument density	Development of instruments over time (difference between number of adopted and abolished instruments)
Intensity change	Intensity level	Development of regulatory instrument strictness over time (difference between number and/or degree measures with increasing and decreasing effects)
	Intensity scope	Development of personal scope/substantive scope/temporal scope of a regulatory instrument over time (difference between number and/or degree measures with increasing and decreasing effects)

The definitions already make clear that we advocate a measurement of regulatory policy change in terms of aggregate developments comprising a large number of changes in state activity within a given policy field.

Table 2 summarizes the proposed dimensions and sub-dimensions of density change and intensity change and attaches a set of indicators to them. The latter are only broadly introduced at this point, but in the next section we outline the operationalization in more detail by means of clean air regulations.

An increase in regulatory density points toward expansion, whereas a decrease in regulatory density can be interpreted as dismantling. Changes in regulatory density can be measured via two variables: the number of regulatory targets and the number of instruments within a policy field. Note that the relationship between the two variables is inherently hierarchical, which is captured by our conception. Thus, whenever a regulatory target is changed, this also entails at least one event of change at the instrument level because a target is always regulated by means of at least one instrument. However, a change in instruments can occur without changes in the regulatory targets. Consequently, while it is never possible to observe one event of change in a regulatory target without an

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event of change in instruments (which are counted as two events of change in regulatory density), it is possible that instruments change while the regulatory target remains unchanged.

It generally holds that the larger the number of regulatory targets, the higher the regulatory penetration of a field is. For example, in order to reduce air pollution, governments can identify a plethora of regulatory policy targets. On the one hand, they can regulate the amount of pollutant emissions. The corresponding targets would then refer to the different substances whose emissions are regulated, for instance, sulfur dioxide (SO_2). On the other hand, governments can also define specific air quality goals with regard to various substances, such as particulate matter (see e.g. Jordan *et al.* 2010). In the aggregate, an increase in the number of regulatory targets in a policy field indicates expansion, whereas dismantling is manifested in a decrease in the number of regulatory targets.

The second variable for measuring regulatory density is determined by the number of instruments in a given policy field. The number of instruments measures regulatory density on a more concrete level than does the number of targets. To reduce the amount of industrial carbon dioxide (CO_2) emissions into the air, for instance, governments can rely upon a broad array and combination of instruments, including command-and-control approaches (the definition of legally binding emission standards), economic incentives (such as environmental taxes or emission trading systems), or industrial self-regulation and voluntary agreements (for an overview, see Sterner 2002). As already mentioned, even as the number of regulatory policy targets in a given policy area remains constant over time, the number of instruments can hence increase or decrease.

We now turn to the dimension of regulatory intensity. Analogous to regulatory density, an increase in intensity over time can be understood as expansion, whereas a decrease indicates dismantling. We measure changes in intensity using two variables: the intensity level and the intensity scope. Both variables refer to the specific calibration of a given regulatory instrument. They measure two separate phenomena, which are combined in case changes occur in both. Changes in intensity levels rest upon potential increases and reductions of the concrete amount of permissible emissions. Changes in intensity scope, by contrast, refer to the number of cases, constellations, or addressees covered by a certain instrument. For instance, the scope of an emission standard regulating certain emissions from combustion plants increases once the threshold defining the size of plants covered by the regulation is lowered. In this case, more companies would be covered by the emission standard and would therefore have to comply with the respective legal obligations.

3. Putting density change and intensity change into practice

In this section, we operationalize our two dimensions of regulatory policy change for the case of clean air regulations. Our operationalization is guided by the main purpose of this study, namely to offer a macro-comparative test of the relationship between regulatory policy outputs and impacts. To this end, the operationalization should facilitate an encompassing measurement and comparison of developments in clean air policymaking across countries and over time. We measure clean air policy change as an aggregate count of all events of change in regulatory policy outputs coded according to

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Table 3 Targets of clean air regulations

Air quality
1. Air quality standard for nitrogen oxides (NO_x)
2. Air quality standard for sulphur dioxide (SO_2)
3. Air quality standard for carbon monoxide (CO)
4. Air quality standard for particulate matter
5. Air quality standard for ozone (O_3)
6. Air quality standard for lead
Emissions (from stationary or mobile sources, product standards)
7. Nitrogen oxide (NO_x) emissions from large combustion plants using coal
8. Nitrogen oxide (NO_x) emissions from passenger vehicles using unleaded gasoline
9. Nitrogen oxide (NO_x) emissions from heavy duty vehicles using diesel
10. Sulphur dioxide (SO_2) emissions from large combustion plants using coal
11. Sulphur dioxide (SO_2) emissions from passenger vehicles using unleaded gasoline
12. Sulphur dioxide (SO_2) emissions from heavy duty vehicles using diesel
13. Carbon dioxide (CO_2) emissions from large combustion plants using coal
14. Carbon dioxide (CO_2) emissions from passenger vehicles using unleaded gasoline
15. Carbon monoxide (CO) emissions from large combustion using coal
16. Carbon monoxide (CO) emissions from passenger vehicles using unleaded gasoline
17. Particulate matter emissions from large combustion plants using coal
18. Arsenic emissions from stationary sources
19. Maximum permissible limit for the lead content of gasoline
20. Maximum permissible limit for the sulphur content of diesel

the proposed measurement concept. The production of this data has involved a massive collection effort of environmental legislation and its coding by trained coders according to a predefined manual. This data has been compiled in the context of the CONSENSUS project and has been subject to extensive cross-checks with existing sources of environmental legislation.[2] Thanks to this comprehensive data set, we are now able to study changes in clean air regulations in 24 OECD countries from 1976 to 2003.

To begin, Table 3 lists all regulatory targets considered here to assess changes in clean air policy. Overall, 20 targets are examined that refer to both air quality and pollutant emissions, as well as product standards. If one of the listed items becomes subject to regulatory intervention, we count it as one event of regulatory expansion. Conversely, if one of the items is not regulated anymore (a rare event, of course), we count this as an event of regulatory dismantling.

Second, we consider every instrument that is used to achieve the underlying regulatory objective of each of the specified targets. The types of instruments examined are listed in Table 4. Note that there can be several instruments of the same type, for example, several technological prescriptions, in place to regulate a given target. Again, the introduction of a new instrument counts as an event of expansion, whereas the abolition of an existing instrument counts as an event of dismantling.

Finally, we count every change in the level and scope of an instrument as an event of either regulatory expansion or dismantling. A more precise way to assess change in regulatory intensity would of course have been to precisely quantify the changes in level

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Table 4 Regulatory policy instruments

Instrument	Description
Obligatory standard	Legally enforceable numerical standard, typically involving a measurement unit, e.g. mg l^{-1}
Prohibition/ban	Total or partial prohibition/ban on certain emissions, activities, products, etc.
Technological prescription	Measure prescribing the use of a specific technology or process
Tax/levy	Tax or levy for a polluting product or activity
Subsidy/tax reduction	Measure by which the state grants a financial advantage to a certain product or activity
Liability scheme	Measure that allocates the costs of environmental damage to those who have caused the damage
Planning instrument	Measure defining areas or times that deserve particular protection
Public investment	Specific public investment
Data collection/monitoring programmes	Specific programme for collecting data
Information-based instrument	Exchange of information between the state and polluters or among polluters
Voluntary instrument	Voluntary agreements or commitments between the state and private actors or among private actors
Permits	Permit to pollute the environment or to produce/import/export/sell environmentally harmful products

and scope. We do not pursue this approach for the moment, however, because it is very difficult to make these changes comparable across instruments and countries over time. For example, we cannot readily compare the levels of taxes with those of emission standards. Another problem in this context is that some jurisdictions prefer to define specific standards for individual industrial sectors, while others prefer to define universal standards that are valid across all sectors. At this point, we thus acknowledge the weakness of our approach, but still deem it, in the presence of the aforementioned problems, conceptually more appropriate to adhere to an aggregate count of all events of change with expansive and dismantling effects. As a result, the final measure of aggregate regulatory policy change is composed of the difference between all expansive and dismantling events of change in the density and intensity dimensions (see Table 1).

Figure 1 illustrates the changes in clean air regulations – as measured by our concepts of density and intensity change – aggregated over all 24 countries in our sample. The dashed line denotes the count of all events of regulatory expansion, whereas the solid line gives the count of dismantling events. The continuously rising lines represent our principal measures of regulatory density and intensity, that is, cumulative change (expansion minus dismantling) over time. Dismantling activities in the regulation of air pollution do indeed exist for our country sample, but are overall the exception. This leads to a continuous increase in overall regulatory density and intensity, that is, regulatory expansion discounted of dismantling. Note that both expansive and dismantling activities in terms of intensity change are more frequent than density changes. This means that clean

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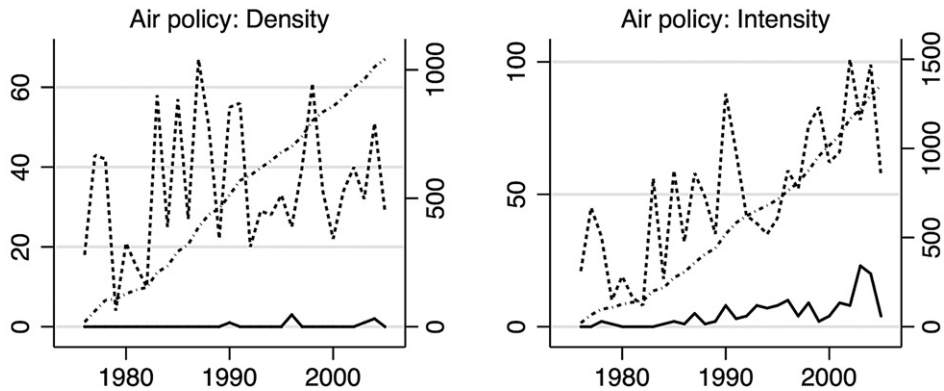

Figure 1 Changes in clean air regulations in 24 OECD countries.

air regulations are more often subject to adjustments in the settings and scope of the respective instruments (i.e. events of intensity change) than affected by changes in the instruments or the adoption or abolition of (new) regulatory objectives (i.e. events of density change).

4. Test of the relationship between air pollution regulations and impacts

We now turn to the key question guiding this study, namely whether changes in regulatory policy outputs can explain changes in regulatory policy impacts. Before we can evaluate this question, however, we first provide details about the dependent and independent variables.

4.1. Introducing the variables

Our dependent variables are national emission intensities, that is, emissions in grams per unit of GDP (see e.g. Cao & Prakash 2010), of three frequently used air pollutants, namely carbon dioxide (CO_2), sulfur dioxide (SO_2) and nitrogen oxides (NO_X). They represent the main air polluting substances employed as dependent variables in numerous empirical studies (see e.g. Roller 2005). The data on the air pollutants' emissions levels was extracted from the Emissions Database for Global Atmospheric Research (EDGAR).[3]

Our core independent variables refer to the effect of regulatory activities measured in terms of clean air policy density (*Regulatory density*) and intensity (*Regulatory intensity*), as discussed in detail in the previous sections. In general, we expect emission intensities to decrease with denser and more intense clean air regulations, and vice versa. Since the two dimensions are highly correlated for clean air regulations, we fit separate models for each of the two analytical dimensions.

We test the effect of clean air regulations on emission intensities against several control variables. First, following the reasoning of the environmental Kuznets curve, demand and supply of environmental policies are expected to increase at higher income levels, which might finally also lead to lower emissions (see e.g. Selden & Song 1994; Grossman & Krueger 1995; Aubourg *et al.* 2008).[4] Moreover, economic downturns can be responsible for reductions in pollution loads (Hughes & Lovei 1999). Contrary to this

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expectation, economic upswings can also boost investment in more advanced and environmentally friendly technologies. We control for these factors by including the natural log of GDP per capita (*GDP* pc) and the growth rate of GDP in our models (*GDP growth*). Apart from these basic economic conditions, the structural composition of national economies should have an impact on emission levels. In particular, the industrial sector can be expected to contribute above average to overall pollution (see e.g. Earnhart & Lizal 2008). We therefore control for the size of the industrial sector via its contribution to total GDP (*Industry*).

Another set of controls refers to the effects of international trade (see e.g. Andonova *et al.* 2007; Holzinger *et al.* 2008, 2011). First, we control for differences in overall *Trade intensity*, measured as exports plus imports as a percentage of GDP. Higher levels of trade can be assumed to either exert downward pressures on environmental standards (Prakash & Potoski 2006) or induce convergence toward higher levels of environmental standards (Vogel 1995). We also use two more specific controls that capture effects emanating from differential patterns of international trade, namely the share of *Manufactures exports* and *Manufactures imports*. Since the production of manufactured goods is known to be particularly pollution intense, the theory of regulatory competition would predict laxer regulation and hence higher emission intensities, particularly for countries that rely more heavily on manufacture exports. The alternative scenario is that increased competition from manufacturing imports and exports has positive effects if domestic firms move toward more environmentally efficient product and production technologies (Perkins & Neumayer 2008). By the same token, beneficial technology spillovers can be expected at higher levels of foreign direct investment (FDI) inflows (Prakash & Potoski 2007). We therefore also include net *FDI inflow* in our models.

Finally, in accordance with previous studies, we control for *Population density* and *Urban population* in order to rule out confounding effects related to demographics. All independent variables – with the important exception of our measurements of *Regulatory density* and *Regulatory intensity*, which are based on original data – are derived from the World Bank's World Development Indicators.[5] The dependent and independent variables are summarized in Table 5.

4.2. Presentation of the key findings

We estimate the relationship between emission intensities and the independent variables by means of standard panel analysis techniques. As is customary in the literature, all independent variables are lagged by one year. We include country fixed effects in our analyses because we are interested in the effects of changes in regulatory policy outputs on environmental impacts, rather than in the effects of differing regulatory levels across countries. In this way, we are also able to control for unobserved country-level heterogeneity and to ensure that the results are not driven by particular countries (Kittel & Winner 2005).

In our first set of models, reported in Table 6, we use panel-corrected standard errors with corrections for first-order autocorrelation to adjust for disturbances arising from cross-sectional heteroscedasticity, contemporaneous correlation, and temporal autocorrelation (Beck & Katz 1995, 2011).

The findings show that regulatory policy outputs have a significant negative effect overall on emission intensities, thus supporting our basic reasoning about the relationship between outputs and impacts. More precisely, it means that higher levels of regula-

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Table 5 Descriptive statistics

Variable	Description	Mean	SD	Min	Max
CO_2	CO_2 emission intensity (grams of emission per unit GDP, log)	17.839	0.415	16.691	18.600
SO_2	SO_2 emission intensity (grams of emission per unit GDP, log)	12.369	0.999	9.631	13.947
NO_X	NO_X emission intensity (grams of emission per unit GDP, log)	12.243	0.611	10.564	13.913
Regulatory density	Clean air policy density	19.875	18.672	0	88
Regulatory intensity	Clean air policy intensity	21.590	23.019	0	114
GDP pc	GDP per capita (2000 US$, log)	9.627	0.595	7.794	10.559
GDP growth	Annual GDP growth (%)	2.930	2.599	−6.854	11.494
Population density	Population density (people per sq km of land area)	133.978	132.285	1.826	494.411
Urban population	Urban population (% of total population)	71.802	11.921	41.2	97.22
Industry	Industrial added value (% of total GDP)	31.768	4.867	19.034	42.623
Trade intensity	Imports + exports (% of GDP)	60.607	29.224	9.102	184.742
Manufactures exports	Manufactures exports (% of merchandise exports)	68.321	21.785	9.361	96.558
Manufactures imports	Manufactures imports (% of merchandise imports)	68.932	12.437	18.315	87.438
FDI inflow	Foreign direct investment net inflows (% of GDP)	2.017	5.073	−0.663	92.498

Notes: Number of observations, 632. Max, maximum values; Min, minimum values; SD, standard deviation.

tory density and intensity are associated with lower levels of emission intensities, suggesting a positive effect of environmental regulatory efforts in the countries under study. Only in the case of CO_2 emissions, the coefficient for regulatory density does not reach conventional levels of statistical significance.

With regard to the control variables, our results confirm most of the discussed expectations based on previous research. While emission intensities for all three pollutants decrease at higher levels of per capita income in a country, periods of strong economic growth increase emission intensities for CO_2 and SO_2. Moreover, as expected, higher shares of value added by industrial production increase emission intensities (except for model 2 where the coefficient is insignificant). Urbanization also has a significant positive effect on emissions, whereas increases in population density result in a significant negative predictor of SO_2 emissions only.

Interestingly, higher levels of overall trade are associated with lower emission intensities for all three pollutants. At the same time, levels of FDI inflow cannot be associated with emission intensities. Together, these findings point to a beneficial effect of economic integration on environmental quality, at least in advanced democracies. With regard to the more specific characteristics of trade in our models, we can only find a significant reduction in SO_2 emissions at higher levels of manufactures imports, whereas exports of manufactures do not seem to have a systematic effect on emission intensities. In

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Table 6 Determinants of air emissions, 1976–2003. Specification in levels

	(1) CO_2	(2) CO_2	(3) SO_2	(4) SO_2	(5) NO_X	(6) NO_X
Regulatory density	−0.0002 (0.000)		−0.0029** (0.001)		−0.0017** (0.001)	
Regulatory intensity		−0.0012*** (0.000)		−0.0055*** (0.001)		−0.0030*** (0.001)
GDP pc	−0.7360*** (0.064)	−0.6844*** (0.063)	−1.7159*** (0.164)	−1.5500*** (0.165)	−1.1387*** (0.115)	−1.0507*** (0.115)
GDP growth	0.0021* (0.001)	0.0019* (0.001)	0.0052** (0.003)	0.0044* (0.002)	0.0018 (0.002)	0.0014 (0.002)
Population density	−0.0019 (0.001)	−0.0017 (0.001)	−0.0109*** (0.003)	−0.0101*** (0.003)	−0.0033 (0.002)	−0.0029 (0.002)
Urban population	0.0261*** (0.004)	0.0250*** (0.004)	0.0446*** (0.013)	0.0406*** (0.012)	0.0388*** (0.007)	0.0362*** (0.008)
Industry	0.0040** (0.002)	0.0026 (0.002)	0.0337*** (0.005)	0.0290*** (0.005)	0.0128*** (0.004)	0.0102*** (0.003)
Trade intensity	−0.0015** (0.001)	−0.0013** (0.001)	−0.0054*** (0.002)	−0.0047*** (0.002)	−0.0025** (0.001)	−0.0021** (0.001)
Manufactures exports	0.0007 (0.001)	0.0007 (0.001)	0.0025 (0.002)	0.0025 (0.002)	−0.0013 (0.002)	−0.0013 (0.002)
Manufactures imports	−0.0012 (0.001)	−0.0011 (0.001)	−0.0044** (0.002)	−0.0038* (0.002)	−0.0005 (0.001)	−0.0003 (0.001)
FDI inflow	0.0004 (0.001)	0.0006 (0.001)	−0.0003 (0.002)	0.0001 (0.002)	−0.0000 (0.001)	0.0002 (0.001)
Constant	23.3096*** (0.555)	22.9283*** (0.566)	25.8002*** (1.362)	24.6267*** (1.434)	20.9237*** (1.045)	20.3431*** (1.021)
R squared	0.999	0.999	0.987	0.987	0.995	0.995
N/Countries	632/24	632/24	632/24	632/24	632/24	632/24

Notes: N, Number of observations. Panel-corrected standard errors in parentheses. Coefficients for country fixed effects are not reported. All independent variables lagged one year. ***$P < 0.01$, **$P < 0.05$, *$P < 0.1$.

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accordance with previous research, this finding could be explained by environmental efficiency-enhancing technology and knowledge spillovers, which should be particularly high for imports of advanced capital goods (see Perkins & Neumayer 2008).

We next check the robustness of our findings by including a lagged dependent variable, which is the widely used alternative to dealing with serial correlation and modeling dynamic processes. By including a lagged dependent variable, we effectively assume that current emission levels can be explained by emission levels in the previous period. Beck and Katz (2011, p. 342) also recommend this procedure for fixed-effects specifications, as it outperforms instrumental variable approaches in time series with more than 20 periods.

The results reported in Table 7 show that regulatory policy density and intensity do not turn out to be significant predictors of emission intensity levels. Instead, our findings suggest that the yearly adjustments in emission intensities modeled by the lagged dependent variable approach can best be explained by per capita income and urbanization. More specifically, increases in per capita income lead to lower emission levels of CO_2 and NO_X, whereas urbanization leads to increases in these emissions. With regard to SO_2 emissions, the coefficients for both variables are not significant. Moreover, increasing shares of industrial production enhance emission levels of SO_2 and NO_X, but not CO_2. The results also moderately support that increasing trade volumes tends to limit CO_2 and SO_2 emissions.

Overall, our lagged dependent variable models cast first doubts on the causal relationship between changes in clean air policies and air pollutant emissions. In particular, periodical adjustments in emission intensities do not seem to follow directly from changes in regulatory activity.

Another problem, further to unit heterogeneity and autocorrelation, which the inclusion of a lagged dependent variable and panel-corrected standard errors alone cannot deal with (see Tables 6 and 7), is non-stationarities. Particularly if the dependent variable is non-stationary, this usually creates an additional source of bias and might lead to falsely confirming relationships that are spurious. As the Dickey-Fuller and Phillips-Perron tests are not very powerful (Maddala & Wu 1999), we use better alternatives, like the Fisher (1963), Levin–Lin (Levin *et al.* 2002), and Im–Pesaran–Shin tests (Im *et al.* 2003) to check for unit roots in our dependent variable. Based on the results (not reported here, but available upon request) we can only reject the existence of a unit root for CO_2. We therefore choose a conservative approach and rely on an additional dynamic specification in first differences (see also Kittel & Winner 2005). Results of the first differences models are reported in Table 8. We do not include country fixed effects, because first-differencing the variables has removed most of the cross-country variation.

The results of the first-difference models confirm that changes in clean air regulations are a significant predictor of changing emission intensities. Still, with the exception of regulatory density in the case of SO_2 emissions, the coefficients of our output measures reveal the expected signs. Less robust are the results for the variables measuring a country's reliance on industrial production and trade, both of which turn out with insignificant coefficients and, for the most part, even with reversed signs.

Judging by the first-difference models, the most robust predictors of emission intensities remain changes in GDP per capita with increases leading to lower emissions and urban population, for which increases are associated with higher emissions. Yet there is also some variation across the different pollutants under study. Growth in per capita

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Table 7 Determinants of air emissions, 1976–2003. Specification in levels including a lagged dependent variable

	(1) CO_2	(2) CO_2	(3) SO_2	(4) SO_2	(5) NO_X	(6) NO_X
Regulatory density	0.0000 (0.000)		0.0002 (0.001)		−0.0002 (0.000)	
Regulatory intensity		−0.0002 (0.000)		−0.0003 (0.000)		−0.0004 (0.000)
GDP pc	−0.0748*** (0.026)	−0.0676*** (0.023)	−0.0967 (0.063)	−0.0827 (0.062)	−0.1622*** (0.049)	−0.1597*** (0.045)
GDP growth	−0.0026** (0.001)	−0.0027** (0.001)	−0.0029 (0.002)	−0.0030 (0.002)	−0.0041** (0.002)	−0.0042** (0.002)
Population density	−0.0008* (0.000)	−0.0008 (0.000)	−0.0024** (0.001)	−0.0024** (0.001)	−0.0005 (0.001)	−0.0004 (0.001)
Urban population	0.0044*** (0.002)	0.0044*** (0.002)	0.0060 (0.005)	0.0060 (0.005)	0.0051** (0.002)	0.0052** (0.002)
Industry	0.0002 (0.001)	−0.0003 (0.001)	0.0059*** (0.002)	0.0051** (0.002)	0.0053*** (0.002)	0.0049*** (0.002)
Trade intensity	−0.0007** (0.000)	−0.0007* (0.000)	−0.0014* (0.001)	−0.0013* (0.001)	−0.0006 (0.001)	−0.0005 (0.001)
Manufactures exports	−0.0001 (0.000)	−0.0001 (0.000)	−0.0004 (0.001)	−0.0004 (0.001)	−0.0006 (0.001)	−0.0007 (0.001)
Manufactures imports	−0.0001 (0.000)	−0.0001 (0.000)	−0.0004 (0.001)	−0.0003 (0.001)	0.0007 (0.001)	0.0007 (0.001)
FDI inflow	0.0004 (0.001)	0.0005 (0.001)	0.0007 (0.001)	0.0009 (0.001)	0.0002 (0.001)	0.0003 (0.001)
Lagged DV	0.8548*** (0.032)	0.8533*** (0.032)	0.9173*** (0.028)	0.9163*** (0.028)	0.8778*** (0.027)	0.8734*** (0.027)
Constant	3.0556*** (0.668)	3.0211*** (0.647)	1.4303* (0.789)	1.3217* (0.803)	2.5790*** (0.616)	2.6144*** (0.571)
R squared	0.986	0.986	0.989	0.989	0.988	0.988
N/Countries	632/24	632/24	632/24	632/24	632/24	632/24

Notes: DV, dependent variable; N, Number of observations. Panel-corrected standard errors in parentheses. Coefficients for country fixed effects are not reported. All independent variables lagged one year. ***$P < 0.01$, **$P < 0.05$, *$P < 0.1$.

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Table 8 Determinants of air emissions, 1976–2003. Specification in first differences

	(1) ΔCO_2	(2) ΔCO_2	(3) ΔSO_2	(4) ΔSO_2	(5) ΔNO_X	(6) ΔNO_X
Δ Regulatory density	−0.0003 (0.001)		0.0005 (0.001)		−0.0001 (0.001)	
Δ Regulatory intensity		−0.0004 (0.000)		−0.0005 (0.001)		−0.0004 (0.001)
Δ GDP pc	−0.4078*** (0.148)	−0.4049*** (0.148)	−0.3225 (0.266)	−0.3146 (0.267)	−0.4805** (0.213)	−0.4772** (0.213)
Δ GDP growth	0.0004 (0.001)	0.0004 (0.001)	−0.0018 (0.002)	−0.0019 (0.002)	−0.0017 (0.002)	−0.0017 (0.002)
Δ Population density	−0.0012 (0.003)	−0.0012 (0.003)	−0.0146** (0.006)	−0.0145** (0.006)	−0.0073 (0.005)	−0.0072 (0.005)
Δ Urban population	0.0151 (0.011)	0.0147 (0.011)	0.0644*** (0.015)	0.0633*** (0.016)	0.0483*** (0.014)	0.0477*** (0.014)
Δ Industry	−0.0006 (0.003)	−0.0006 (0.003)	−0.0016 (0.004)	−0.0016 (0.004)	−0.0055 (0.004)	−0.0055 (0.004)
Δ Trade intensity	−0.0003 (0.001)	−0.0003 (0.001)	0.0007 (0.001)	0.0007 (0.001)	0.0005 (0.001)	0.0005 (0.001)
Δ Manufactures exports	0.0005 (0.001)	0.0005 (0.001)	0.0017 (0.002)	0.0017 (0.002)	−0.0008 (0.002)	−0.0008 (0.002)
Δ Manufactures imports	−0.0007 (0.001)	−0.0007 (0.001)	0.0013 (0.002)	0.0013 (0.002)	0.0011 (0.001)	0.0011 (0.001)
Δ FDI inflow	0.0006 (0.001)	0.0006 (0.001)	−0.0005 (0.001)	−0.0004 (0.001)	−0.0000 (0.001)	0.0000 (0.001)
Lagged DV	−0.0022 (0.088)	−0.0014 (0.088)	−0.0526 (0.060)	−0.0505 (0.060)	−0.1044 (0.081)	−0.1047 (0.081)
Constant	−0.0105* (0.006)	−0.0102* (0.006)	−0.0698*** (0.011)	−0.0682*** (0.011)	−0.0345*** (0.009)	−0.0338*** (0.009)
R squared	0.118	0.118	0.103	0.103	0.105	0.105
N/Countries	607/24	607/24	607/24	607/24	607/24	607/24

Notes: DV, dependent variable; N, Number of observations. Panel-corrected standard errors in parentheses. All independent variables lagged one year. ***$P < 0.01$, **$P < 0.05$, *$P < 0.1$.

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income is not a significant predictor of SO_2 emission intensities anymore, while urbanization does not have a significant effect on CO_2 emissions. These differences suggest that it may well matter which particular pollutant is studied for conclusions reached by impact data.

In summary, the analyses performed suggest that clean air regulations cannot be associated with environmental impacts under all circumstances. The relationship between outputs and impacts is particularly strong if we rely on absolute emissions levels. However, as soon as we eliminate the time trend, the relationship becomes significantly weaker. Put differently, increases in clean air regulatory policy density and intensity do not translate directly into decreases in emission intensities. This observation has the following broader research implications: It corroborates the key insight of implementation research, namely that there is a slippage between regulatory policy outputs and impacts. This slippage may be caused by various forms of implementation problems, as well as by other factors that cannot be adequately measured. However, on the basis of this study we cannot make any valid statements about the exact nature of the factors weakening the relationship between regulatory policy outputs and impacts.

5. Concluding remarks

In this study we pursued the objective of empirically testing the relationship between changes in regulatory outputs and impacts in the field of clean air policy in advanced democracies. While data on clean air policy impacts, typically emissions levels of air pollutants, are collected and provided for analysis by international organizations, we had to develop a new concept for measuring regulatory policy outputs. To this end, we introduced the dimensions of regulatory density and regulatory intensity. The dimension of regulatory density is composed of the regulatory targets and instruments that populate a given policy field. It measures the regulatory penetration and internal differentiation of the field. In a complementary vein, regulatory intensity refers to the strictness of these measures by considering the concrete calibrations of instruments in terms of their levels and scopes of application.

In our view, the dimensions of regulatory density and intensity are appropriate for characterizing regulatory policy change in a comprehensive yet differentiated way. However, if one wishes to have a more nuanced measure and to lower the risk of messy findings, our measurement approach can also be easily modified by disaggregating the two variables composing density change (i.e. regulatory targets and instruments) and intensity change (i.e. the levels and scopes of application of regulatory instruments). By the same token, it would be possible to sum up regulatory targets and instruments, as well as their levels and scope, to produce a single compound measure of regulatory policy change. This flexibility is provided by our measurement approach.

We illustrated the application of our measurement concept by focusing on clean air policy. In a second step, we evaluated whether clean air regulations can explain emission intensities of widely examined pollutants. The results of our analysis show that clean air regulations have no robust effect on emission levels of major air pollutants. In our analyses, the relationship between regulatory policy outputs and impacts vanishes once we eliminate the time trend. Thus, the results of the first model suggest a relationship that must be judged spurious. As a consequence, we conclude that environmental impacts are not unconditionally a reliable proxy for environment policy change, which is sometimes

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implicitly claimed in the literature (see e.g. Wälti 2004; Bernauer & Koubi 2009). Instead, our findings rather support the main argument of the policy implementation literature, which claims that regulatory outputs do not automatically lead to the desired impacts (see e.g. Pressman & Wildavsky 1973; Pülzl & Treib 2007; Knill & Tosun 2012, ch. 7). While properly relating our findings to implementation research would go beyond the scope of this paper, we expect that this literature could be helpful in determining to which degree they are generalizable to other areas of regulatory policymaking. In this context, our findings become even more relevant when recognizing that we analyzed this relationship for advanced democracies in which responsible government agencies usually have greater willingness and capacity to monitor and enforce compliance with regulatory standards. Thus, the correlation between regulatory policy outputs and impacts may be even weaker in less advanced countries (see e.g. Andonova *et al.* 2007). Consequently, we encourage future research to pay more attention to the reported slippage between regulatory policy outputs and impacts. To this end, a more intensive exchange between academics and practitioners might prove to be a seminal strategy.

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Notes

1 The data were collected in the context of the collaborative project CONSENSUS. For more detailed information visit http://www.fp7-consensus.eu.
2 The data collection involved over 30 country experts that had to participate in various coders' meetings to train them appropriately. The coding of the data was the exclusive responsibility of the project teams located in six European universities in order to ensure a maximum level of consistency. The data had to pass through tests of inter-coder reliability, as well as extensive cross-checks with existing data sources of clean air regulations in the countries under study.
3 Available from URL: http://edgar.jrc.ec.europa.eu/.
4 Note that the Environmental Kuznets Curve only models the relationship between income and pollution intensity. It does not take into account, for example, the effects of higher levels of consumption because of an increase in income for pollution intensity.
5 Available from URL: http://data.worldbank.org/indicator.

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